LANGUAGE :

It is a medium to communicate between two entities

PROGRAMMING LANGUAGE:

A language or medium which is used to communicate or instruct a computer to perform a specific task is known as programming language.

Different types of programming language:

There are 3 types of programming language.

They are:

1. Machine level/Low level language

2. Assembly level/Mid-level language

3. High level language

1. Machine level/Low level language

A language understandable, readable and executable by a machine is known as machine level language. E.g.: Binary language (consist of 0 and 1).

This language is quite easy and friendly for machines but very difficult for humans to understand.

2. Assembly level/Mid-level language

Assembly level language consists of pre-defined words, and these words are known as instruction sets or mnemonics.

We have 8086 instruction sets.

E.g : ADD (Which is used to add to numbers) MOV (To move data from one memory location to another memory location) SUB (Used to subtract two numbers)

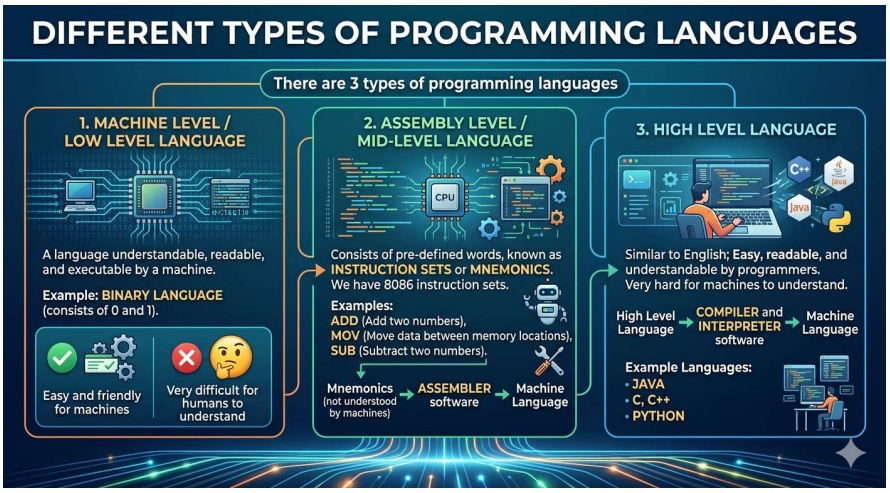
These mnemonics will not be understandable by the machine, so we have to convert it into machine understandable language with the help of software known as ASSEMBLER

3. High level language

Language which is similar to English and it is easy, readable and understandable by the programmers. But very hard for the machines to understand.

High level language is converted to machine language with the help of software's such as COMPILER and INTERPRETER.

E.g.: JAVA, C, C++, python.



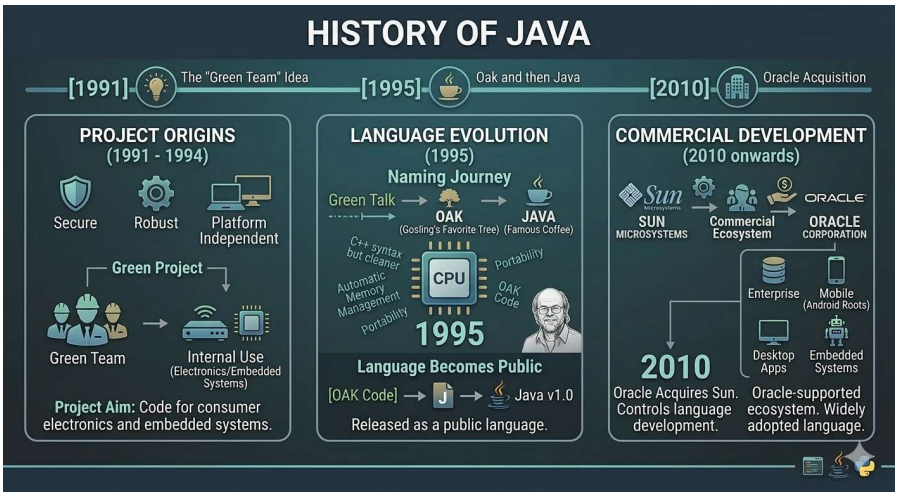
History of Java :

James Gosling invented java

AIM: To create a language which is robust, platform independent, secure which will be used to communicate or program with electronics, embedded system, set-up boxes.

1. Group of engineers worked together is Green team.
2. Name of the project: Green project

3. Name of language first created is Green talk then changed into oak (fond of oak tree by James Gosling) then to Java (a famous coffee during that time).
4. Company Name: Sun Microsystem.
5. It started in the year 1991 as an internal project.
6. By 1995 it became a public project.
7. Around 2010 Oracle acquired the Sun Microsystem and now the project is under oracle.



Features of JAVA

Easy Way to Remember (Exam Trick)

Remember this line:

"SOP SPHMRDD"

- **S** – Simple
- **O** – Object Oriented

- **P** – Platform Independent
- **S** – Secure
- **P** – Portable
- **H** – High Performance
- **M** – Multithreaded
- **R** – Robust
- **D** – Distributed
- **D** – Dynamic